IMPLEMENTING REINFORCEMENT Q-LEARNING IN TRAFIC SIGNAL OPTIMIZATION FOR ONE INTERSECTION

(Nghiên cứu ứng dụng học tăng cường Q-learning trong việc điều chỉnh tín hiệu đèn giao thông tại một giao lộ)

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Abstract

As the population increases, urban intersections are becoming critical nodes within modern traffic systems and play an important role in the logistics and supply networks. However, these pivotal hubs are often plagued by congestion and inefficiency. Mostly controlled by fixed-cycle programs, these traditional traffic signals lack the adaptability to accommodate dynamic changes in traffic flow, leading to increased emissions, delays, and wasted fuel - which directly impact to delivery costs and effcientcy. For these aforementioned reasons, growing the need for an adaptive traffic light control models that can quickly respond to real-time traffic changes. As artificial intelligence advances in every daily aspects, its applications in smart traffic management brings hope for a new and modern solution to this urban problem.

This study presents a Q-learning-based approach for optimizing traffic signal durations at a four-way intersection. By leveraging reinforcement learning, the model learns to adjust traffic light timings dynamically to balance traffic density across all lanes, prioritize congested roads, and maximize overall throughput. The Q-learning based agent will learn through interacting actions with its environment and using a Q-table to store and update calculated Q-values. The agent can then take actions that either maximize cumulative rewards over time or satisfy any specified requirement. A simulation environment using SUMO (Simulation of Urban Mobility) evaluates the proposed model against traditional fixed-timing systems. Results demonstrate significant improvements in average speed, congestion levels, and waiting times. The findings highlight the potential of Q-learning in traffic management, contributing to smarter and more sustainable flow of goods as well as urban mobility solutions.

Keywords: *Reinforcement learning, Q-learning, Traffic light control, Intelligent transportation system, Urban mobility.*

1. Introduction

As the population increases day by day, a complex and pressing challenge is rising in every urban area, which is traffic signal optimization. As most traffic lights are still controlled by fixed-cycle systems operating on predefined schedules, this causes the infrastructure to lose its flexibility and cannot adapt to sudden changes in traffic flow leading to various disadvantages, including increased air and noise pollution and wasted fuel.

For these aforementioned reasons, growing the need for adaptive traffic light control models that can quickly respond to real-time traffic changes. As artificial intelligence advances in every daily aspect, its applications in smart traffic management brings hope for a new and modern solution to this urban problem.

Having the ability to learn through interactions with the environment, Q-learning techniques within Reinforcement learning (RL) stands out for its effectiveness. Continuously learning from the rewards or penalties of each decision, the Q-learning agent can identify the best actions for various scenarios and easily adjust different actions based on input traffic volume [5]. Realizing its potential, many academic researches have been done internationally. A 2022 review emphasized the benefits of reinforcement learning in urban traffic management, noting a significant increase in research output in this field - from 5 studies in 2012 to 27 in 2019 - illustrating the growing interest and efficacy of RL in tackling urban congestion [4].

Inheriting insights from previous studies and addressing fixed-cycle system's limitations, this research aims to offering a practical Q-learning-based approach for optimizing traffic signal durations to reduce congestion and improve urban traffic flow in real-time. By designing a reward function as well as two different policies, the proposed model learns the impact of its decisions in many circumstances, achieving more precise control over traffic flow. Using SUMO (Simulation of Urban Mobility) as the simulation environment, this study evaluates the model's performance in realistic scenarios, focusing on factors such as congestion, average speed on each road as well as waiting times. The structure of this paper is as follows: Section 2 outlines the methodology, including data description and problem formulation; Section 3 details the simulation environment and Q-learning implementation; Section 4 presents the results, analysis, and discussion; and Section 5 concludes the paper with future research directions.

2. Experiments

2.1 Q-learning

As a simple yet essential technique in reinforcement learning for any approach, Q-learning is well-known for its adaptability. By creating an agent to learn through interacting actions with its environment and using a Q-table to store and update calculated Q-values, the agent can then take actions that either maximize cumulative rewards over time or satisfy any specific requirement. Through a trial-and-error process, the agent takes actions and then receives its rewards or penalties, and updates the Q-values according to this equation:

$$Q^t(s,a) = (1-\alpha)Q^{t-1}(s,a) + \alpha(R^t + \gamma\, max_a Q^{t-1}(s,a))$$

- $Q^t(s,a)$: The updated Q-value for state s and action a at time step t.
- $Q^{t-1}(s,a)$: The previous Q-value for state s and action a at time step $t-1$.
- α : The learning rate, $\alpha \in (0;\, 1]$, which determines the weight of new knowledge compared to existing information.
- R^t : The total reward received after taking action a in state s at time step t.
- γ : The discount factor, $\gamma \in (0;\, 1]$, which determines the importance of future rewards.
- $max_a Q^{t-1}(s,a)$: The maximum Q-value of all possible actions in the next state, representing the best future reward achievable.
- s : Current state of the environment.
- a : Action taken at the current state.
- t : Time step.

A Q-table includes columns as actions and rows as states at different time steps. As Q-learning interacts with the environment, the agent will choose the action to apply according to the policy, the simulated environment will then measure the effect of it. After that, the agent can calculate the reward and Q-value, update the Q-table and move on to the next state [6]. The following pseudo code describes how Q-learning works:

Algorithm 1 Q-Learning Algorithm

Require: Step size $\alpha \in (0;\, 1]$, discount $\gamma \in (0;\, 1]$, initialized $Q(s,a)$ for all $s \in S, a \in A(s)$

for each time steps **do**

Choose A from $A(s_t)$ using a policy derived from Q (e.g., explore, exploit, min-duration)

Take action A, observe reward R_t, and next state s_{t+1}

Update $Q(s,a) \leftarrow Q(s,a) + \alpha[R_t + \gamma max_{a'} Q(s_{t+1}, a') - Q(s,a)]$

$s_t \leftarrow s_t + 1$

end for

Over continuous training, some actions will gradually become higher than others, the agent will identify them as the best actions to take in different scenarios. This makes the algorithm gain its popularity for being able to make environment-based decisions in many dynamic systems, such as traffic management.

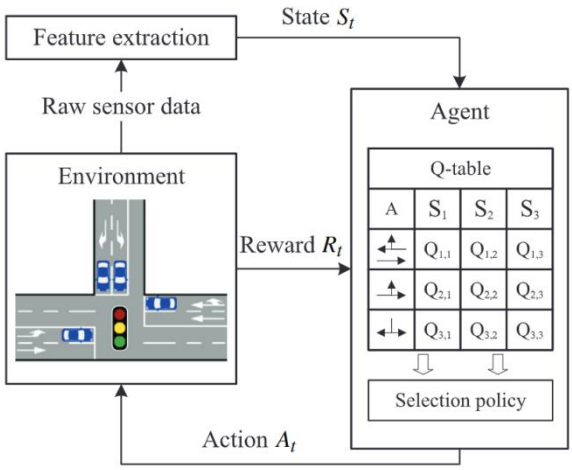


Fig. 1. *Interaction between environment and agent using Q-learning.*

2.2 Components

2.2.1 Simulated Environment

Fig. 2 illustrates the simulated environment of a four-way intersection, including two traffic lanes on each road. Traffic signals are placed at the stop lines of each approach and controlled by the agent. From one approach, the left lane allows vehicles to move straight forward, turn left, or make a U-turn. The right lane permits vehicles to go forward and turn right when the light is in the green phase, as depicted in Fig. 3. In this

research, the agent is designed to control the green and red phase of each approach in order to reduce the congestion level of the intersection. It is worth noting that although this work focuses on a four-way intersection, our solution is promising and can be applied to intersections of any shape.

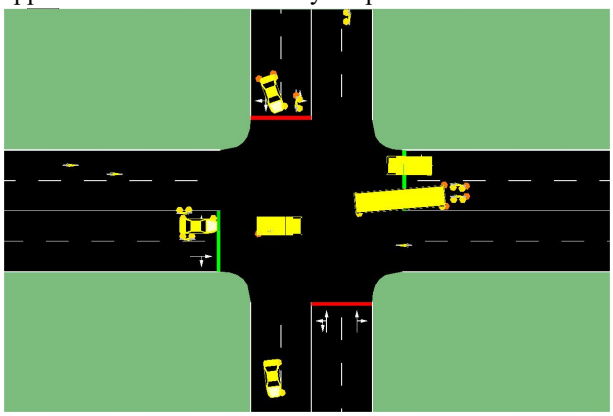


Fig. 2. *Four-way intersection to simulate.*

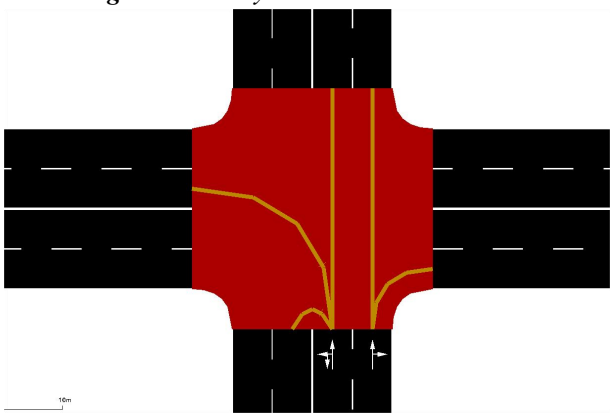


Fig. 3. *Possible direction from one approach.*

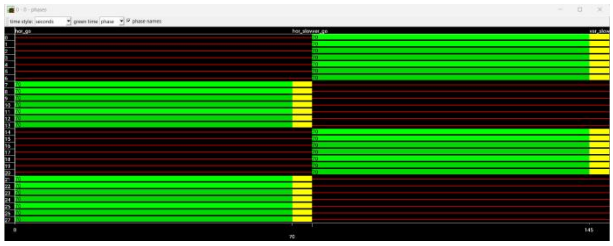


Fig. 4. *Traffic signal phase timing.*

2.2.3 Policy π

At the same time, in different circumstances, the agent's decision could be very based on what policy π driving the agent. Policy π can be understood as the quest that agent needs to accomplish. In traffic control using Q-learning, two popular policies, maximum long-term reward and minimum light duration, stand out among the others. The maximum reward policy π^* aims to actions with highest Q-values, aiming to maximize the cumulative reward. In this policy, the requirement is to maintain the congestion level CL equal or below the desired target $CL_{desired}$. As the reward function is designed to enhance the traffic status through decreasing the congestion level, choosing the most effective actions enables the agent to make timing decisions to solve congestion during rush hours.

On the other hand, the minimum duration policy π_{min} is better suited for off-peak hours when the traffic volume is generally low. By minimizing the vehicle's stop time, it prevents commuters from unnecessary waiting, maintains the highest possible average speed and improves their experience. Together, these policies provide a flexible way to effectively manage the intersection in different cases.

2.2.4. State s

For reward calculation and evaluation, we use average traffic volume in the passenger car unit $\overline{PCU}$, average travel speed $\bar{v}\ (m/s)$, average waiting time $\overline{t_w}\ (s)$ and average congestion level $\overline{CL}\ (\%)$ of each traffic direction. By allowing more vehicles to pass through the intersection and increasing their speed while reducing delay, the intersection's congestion will be addressed, leading to smoother traffic flow and improving overall traffic performance.

The total PCU of one direction at a given time is the sum of all vehicles converted to their respective PCU factors. In this research, PCU factors are considered as per TCVN 4054:2005 (Highway - Specifications for Design) [4], as shown below:

Table 1. *PCU factors used in this study.*

Vehicle Type	Equivalent PCU Factors
Bicycle	0.2
Motorbike	0.3
Car	1.0
Bus	2.0
Regular Truck	2.5

The congestion level at a specific time can also be calculated by using this equation:

$$CL(\%) = \left(\frac{\sum PCU}{C}\right) \times 100$$

with C is the maximum capacity PCU of the approach. [10]

Between two specific time steps, all the average parameters can be calculated as, e.g.:

$$\bar{v}(m/s) = \frac{\sum_i^n \bar{v}_i}{n \times \Delta_t}$$

With Δ_t is the sampling time s, n is the number of samples.

2.2.5. Action

To adjust the traffic light duration, the agent is designed to either increase or decrease the time length with various amounts. The agent is initialized with a resolution $R_t\ (s)$ and core action N_a, indicating how many resolution units the agent can increase or decrease

the duration at a time. As the resolution decreases, the agent can apply even the smallest changes to improve the overall status. Meanwhile, increasing the core action allows the agent to take more effective actions, quickly adapting to rapid changes in traffic flow.

2.2.6. Reward

Compared to being the soul of Q-learning, the reward function is the most important component. It encourages the agent to follow the intended policy addressing the problem or satisfying any requirements. By providing higher rewards, it guides the agent on what actions should and shouldn't be taken in unique situations. In any intersection, it can also be considered as a crossing of many roads. One road will be determined as the main road, characterized by the highest volume or total PCU, with the greatest contribution proportion P_{main} and the sub-roads with P_{sub}. The proportion follows this formula:

$$P_i = \frac{\overline{PCU_i}}{\sum_j \overline{PCU_j}}$$

In cases when all roads have the same proportion, the first road will be considered the main road.

Let R^t_{inst} be the instantaneous reward, defined immediately after taking action at time t, and let R^t_{delay} be the delay reward at time t, which can only be measured and calculated at time $t+1$. The reward function can be expressed as:

$$R_t = R^t_{inst} + R^t_{delay}$$

Since the min-duration policy only focuses on actions with the highest decrement in light duration without violating the $CL_{desired}$, the instantaneous reward is calculated as follows:

$$R^t_{inst} = \frac{\Delta d_{main} - \Delta d_{sub}}{N_a \times R_e} \times 100$$

with Δd_{main} and Δd_{sub} represent the adjusted duration amounts for the main and sub-road, respectively. When increasing the green phase of the main road and the red phase of the sub-road is accordingly extended, this mechanism redistributes the traffic volume within the intersection. The system will continue this adjustment till its volume is below the desired threshold.

Also, two factors commonly used to form a reward function for controlling and improving traffic flow are congestion level and average travel speed. Focusing on these two aspects, we designed a delay reward function to smooth and enable a faster traffic flow:

$$R^t_{delay} = \sum_i P_i \times \left(\frac{\Delta \overline{CL}}{\overline{CL_t}} - \frac{\Delta \overline{v}}{\overline{v_t}} \right) \times 100$$

The greater the decrease in congestion level and the increase in travel speed, the optimal that action is.

2.2.7 Implementation

In each training episode, the agent interacts with the environment. At time step t, state s_t is stored as the current state. The agent then takes all the possible actions, respectively and records these as observed states. After measuring their effects, the best action will be chosen to apply, moving to the next state s_{t+1}. Our training process can be modeled as:

Algorithm 2 Agent Training Algorithm

Require: Step size $\alpha \in (0; 1]$, discount $\gamma \in (0; 1]$, initialized $Q(s, a)$ for all $s \in S, a \in A(s)$
  while episode not terminated **do**
    Initialize s_t
    for each action $a \in A(s_t)$ **do**
      Take action a, observe reward R^t, and next state s_{t+1}
      Update $Q(s_t, a) \leftarrow Q(s_t, a) + \alpha[R^t + \gamma max_{a'} Q(s_{t+1}, a') - Q(s_t, a)]$
    end for
    Choose a_t based on policy $\pi(s_t)$ (e.g., max-reward)
    Transition to next state: $s_t \leftarrow s_{t+1}$
  end while

3. Results and Discussion

The results section focuses on evaluating the traffic conditions at a four-way intersection with two lanes: one main lane (horizontal) and one auxiliary lane (vertical). The yellow line indicates the traffic statistics before applying the Q-learning method (observed) while the blue line indicates the traffic statistics after applying the Q-learning method (validation).

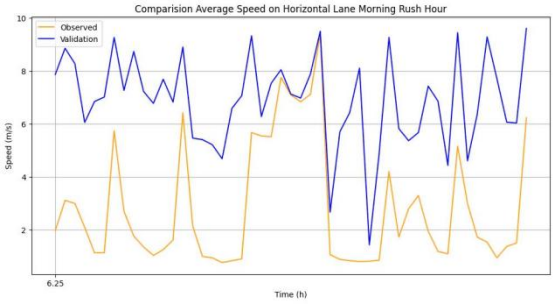


Fig 5. *Average speed on horizontal lane morning rush hour* (m/s).

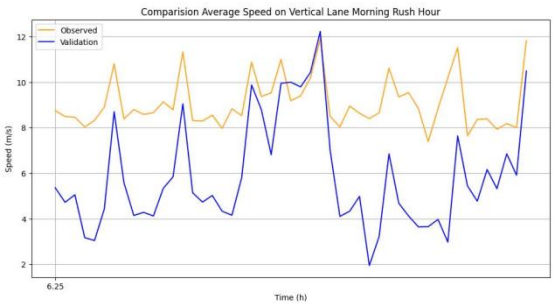


Fig 6. *Average speed on vertical lane morning rush hour* (m/s).

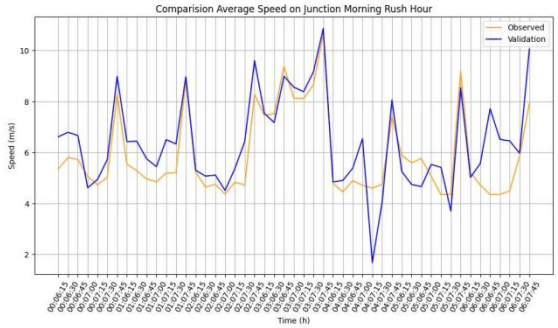


Fig 7. *Junction's average speed on morning rush hour* (m/s).

Table 2. *Mean of recorded average speed morning rush hour* (m/s).

	Vertical road	Horizontal road	Junction
Observed	9.084	2.829	5.902
Validation	5.876	6.890	6.383
Improvement (%)	−35.317	143.500	8.154

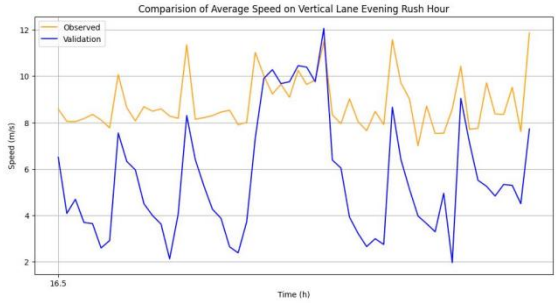


Fig 8. *Average speed on vertical lane evening rush hour* (m/s).

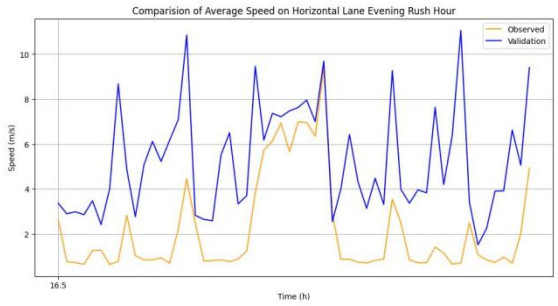


Fig 9. *Average speed on horizontal lane evening rush hour horizontal line* (m/s).

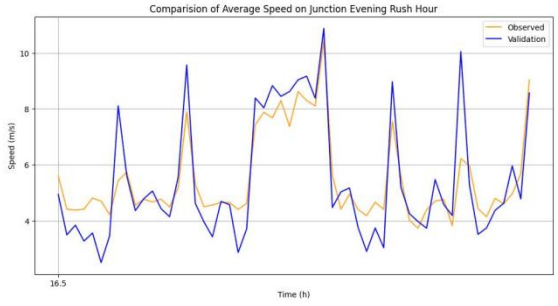


Fig 10. *Average speed on junction evening rush hour.*

Table 3. *Mean of recorded average speed evening rush hour* (m/s).

	Vertical road	Horizontal road	Junction
Observed	8.813	2.186	5.498
Validation	5.593	5.252	5.422
Improvement (%)	−36.535	140.297	−1.368

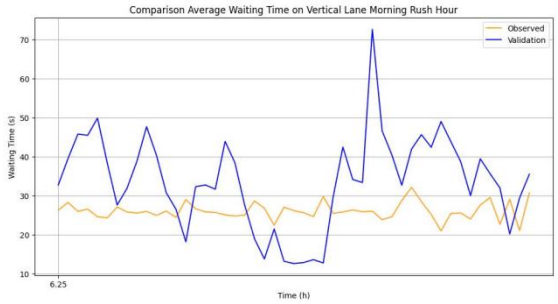


Fig 11. *Average waiting time on vertical morning rush hour* (s).

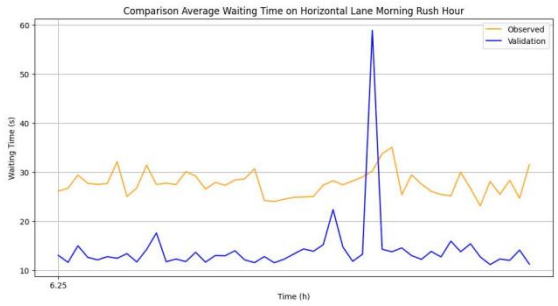


Fig 12. *Average waiting time on horizontal morning rush hour* (s).

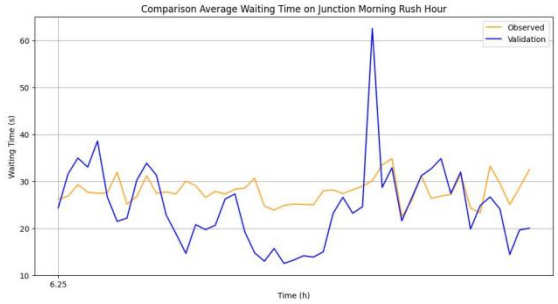


Fig 13. *Average waiting time on junction morning rush hour* (s).

Table 4. *Mean of recorded average waiting time morning rush hour* (s).

	Vertical road	Horizontal road	Junction
Observed	26.140	27.674	29.363
Validation	33.816	14.278	24.627
Improvement (%)	29.363	−48.407	−11.625

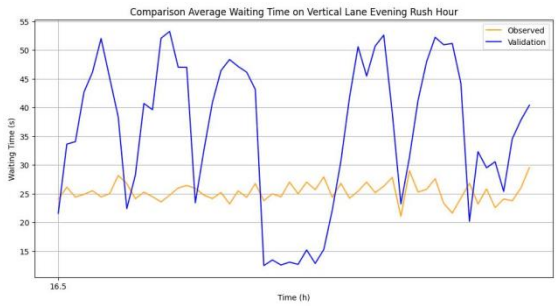


Fig 14. *Average waiting time on vertical evening rush hour* (s).

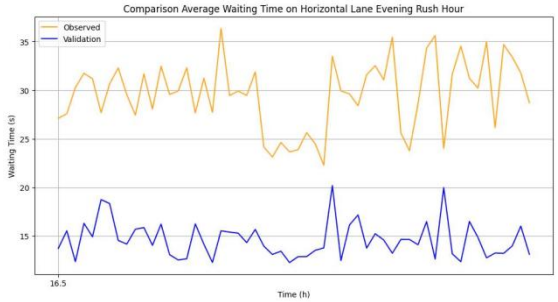


Fig 15. *Average waiting time on horizontal evening rush hour* (s).

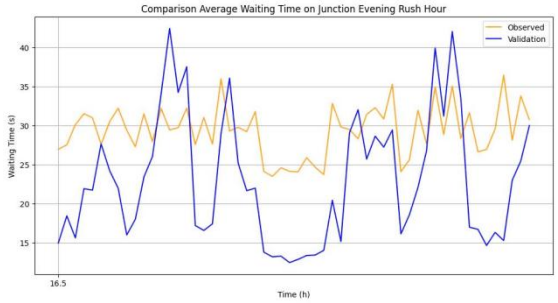


Fig 17. *Average waiting time on junction evening rush hour* (s).

Table 5. *Mean of recorded average waiting time evening rush hour* (s).

	Vertical road	Horizontal road	Junction
Observed	25.286	29.583	29.376
Validation	35.814	14.619	22.961
Improvement (%)	41.635	−50.583	−21.840

These statistics show that the agent can reduce congestion by redistributing traffic between lanes, improving conditions within the intersection while limiting its impact to that area.

4. Conclusions

The experimental results demonstrate that the proposed Q-learning approach effectively controls traffic light durations by adapting to real-time conditions and leveraging past experiences, leading to improved traffic flow, reduced congestion, and better junction performance based on features like average speed, waiting time, and vehicle halts. While the method outperforms traditional fixed-timing systems, it currently lacks consideration of spatial dependencies between intersections. Future improvements could involve integrating multi-source real-time data and developing more adaptive models. Nonetheless, the approach shows strong potential for intelligent traffic management and supports informed decision-making by traffic authorities.

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